



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name: **VESPER**

Team Leader Name: **Ansul Suryawanshi**

Problem Statement: **PS7 — AI-enabled Detection of Exoplanets from Noisy Astronomical Light Curves**

VESPER — Validation Engine for Stellar Photometric Evidence & Recovery

Team Members

Team Leader

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Samiksha Choudhary

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Team Member-3

— optional (BAH allows a team of 3–4)

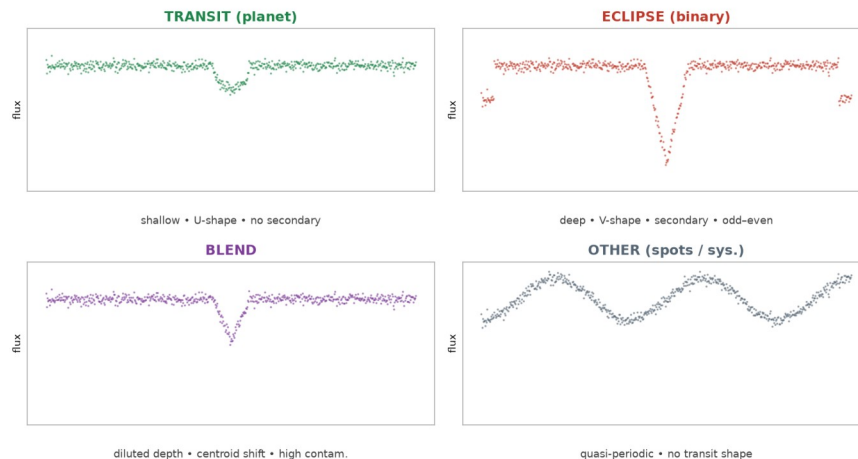
Opportunity

- **Different from existing ideas** — most pipelines either detect OR classify, often with uncalibrated scores. TLS finds a periodic dip; VESPER goes further — it says WHAT the dip is (transit / eclipse / blend / other) with calibrated confidence, on a spine whose recall is non-inferior to full TLS.
- **How it solves the problem** — detect periodic dips → classify with physics + shape discriminators → attach SNR + calibrated confidence → fit period / depth / duration → visualise.
- **USP** — hybrid physics + deep learning (interpretable, small-data robust); calibrated confidence (conformal); physics decides detection (recall-first); light-curve blend tells now, pixel-level centroid / difference-imaging in Round-2; extends a validated, pre-registered spine.
- **Already demonstrated on fresh MAST data** — Pi Mensae c blindly recovered ($P = 6.262$ d, 289 ppm, SDE 12.3); TIC 100029948 flagged as an eclipsing binary (depth ~25%, odd–even 0.23).

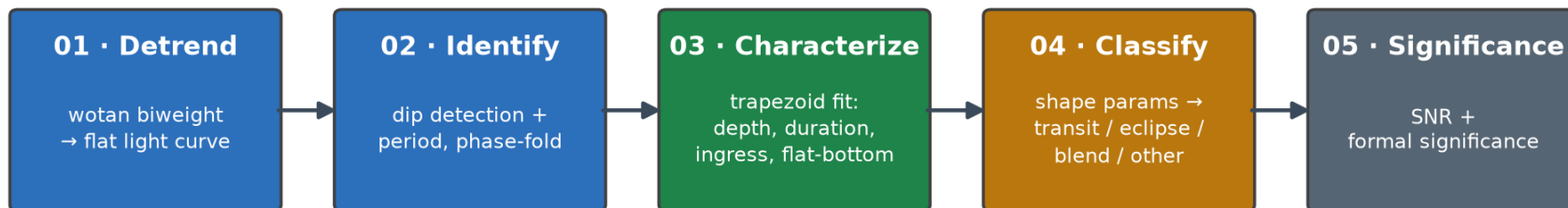
Features

- **Robust periodic-dip detection** — in noisy, crowded fields
- **4-class AI classification** — transit / eclipse / blend / other (physics features now; + CNN in Round-2)
- **SNR / significance** — for every detected event
- **Transit parameter fit** — period, depth, duration (+ uncertainties)
- **Calibrated confidence** — + annotated visualisations
- **Reproducible** — recall non-inferior to full TLS

Four signal classes — distinct physical signatures the classifier separates



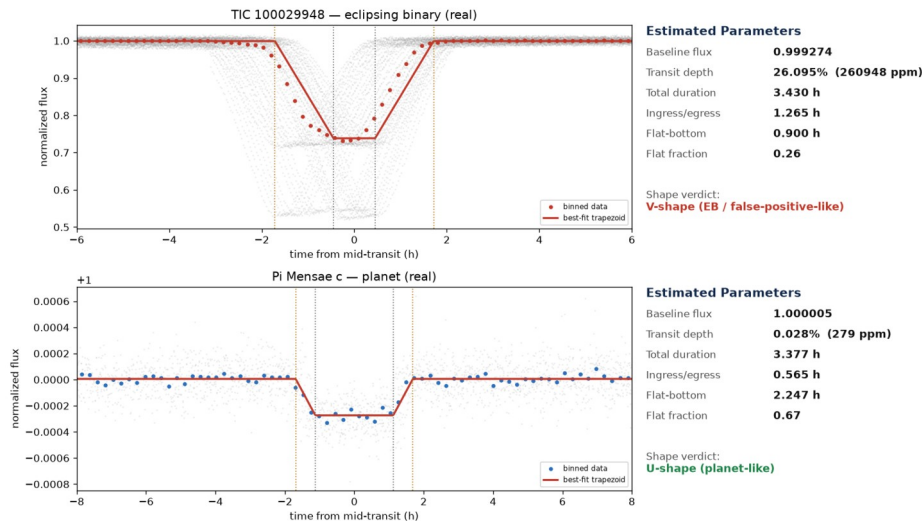
Process Flow — mirrors the PS7 five-step methodology



Steps 01-02 reuse the validated VESPER spine (benchmarked vs full TLS). Step 03 shape-fit feeds Step 04 — **the committee's stated approach: "build the AI classifier on the transit shape parameters."**

Characterization — trapezoid shape fit (PS7 step 03)

Step 03 — Characterization by trapezoid shape fit (depth is equal; SHAPE separates planet from EB)



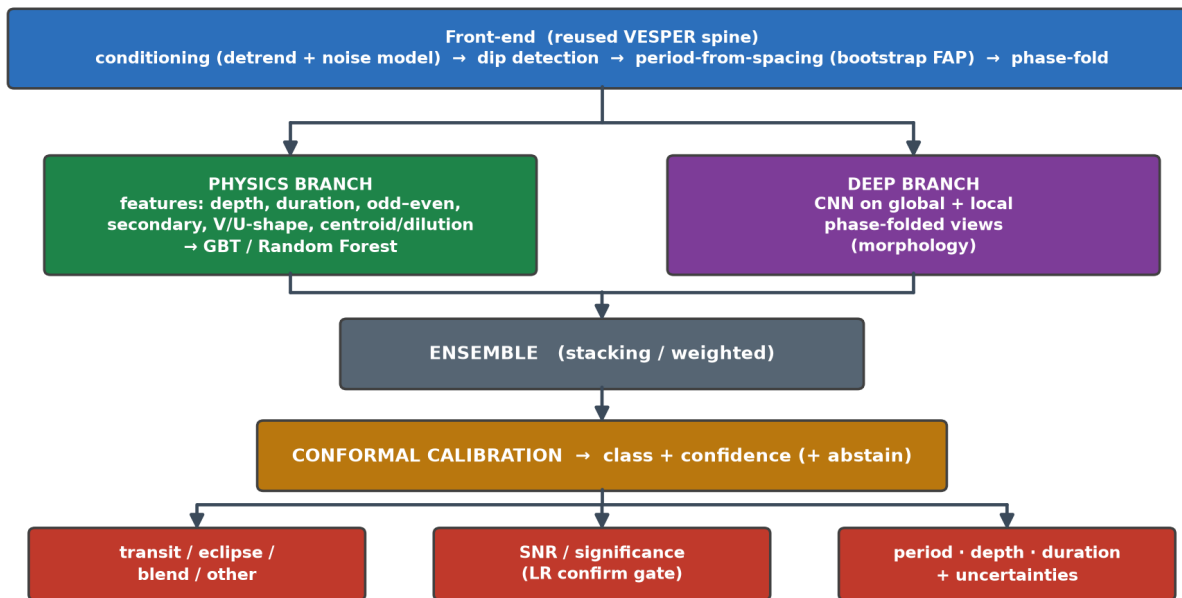
Depth is nearly equal for the EB and the planet — the trapezoid SHAPE separates them:

EB flat-fraction 0.26 (V-shape → false positive) vs planet flat-fraction 0.67 (U-shape) — on real MAST data.

Depth → planet radius (planet vs massive planet); shape U vs V → planet vs eclipsing binary.

Architecture

VESPER — Hybrid Detection & Classification Pipeline



Technologies

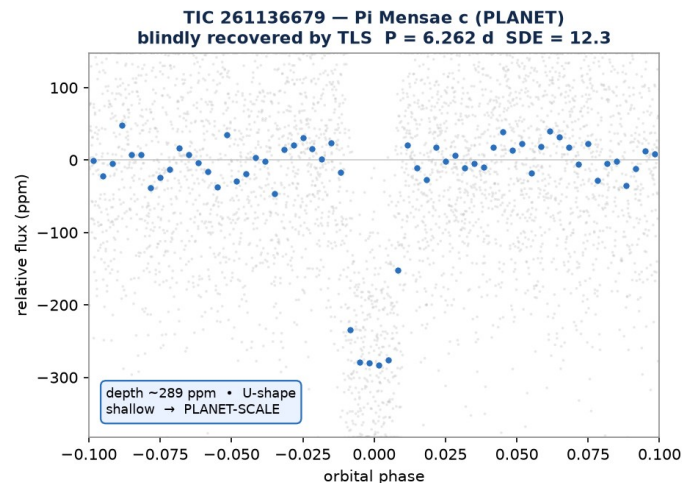
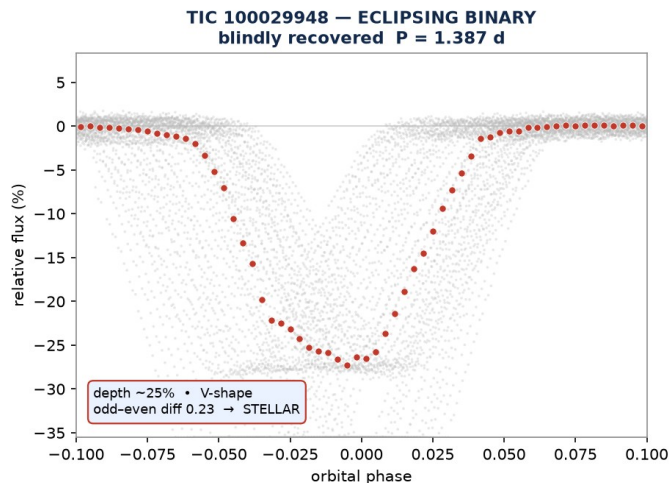
- **Core scientific Python** — numpy · scipy · pandas · matplotlib
- **Astronomy** — lightkurve · astropy · astroquery / MAST · wotan · transitleastsquares · batman
- **Machine learning** — scikit-learn HistGBT (physics branch, built) · PyTorch CNN + conformal / MAPIE (Round-2)
- **Method** — injection–recovery into real LCs · leakage-safe group CV · bootstrap CIs · recall parity vs full TLS
- **All open-source** — no specialised or licensed software; data is public (MAST)

Estimated Implementation Cost

- **Software: ₹0** — entirely open-source stack
- **Data: ₹0** — public TESS archive (MAST); one sector ~40–55 GB raw, kept slim by condition-then-discard
- **Compute (TESS)** — standard laptops for development + a GPU session for CNN training; optional free cloud credits
- **Scales to Kepler / K2 (est.)** — ~200k long-cadence stars; a full transit search $\approx$ 5,000–10,000 CPU-core-hours — a short cloud burst on MAST / AWS Open Data (no egress). Data still ₹0; light curves ~few-hundred GB.
- **Net** — TESS negligible, Kepler a modest cloud burst — feasibility is high; the detection spine already runs on real data

Proof of Concept — fresh MAST data (validated 2026-06-26)

Transit vs Eclipse — fresh MAST data → conditioning → recovery → physics discrimination (PS7)

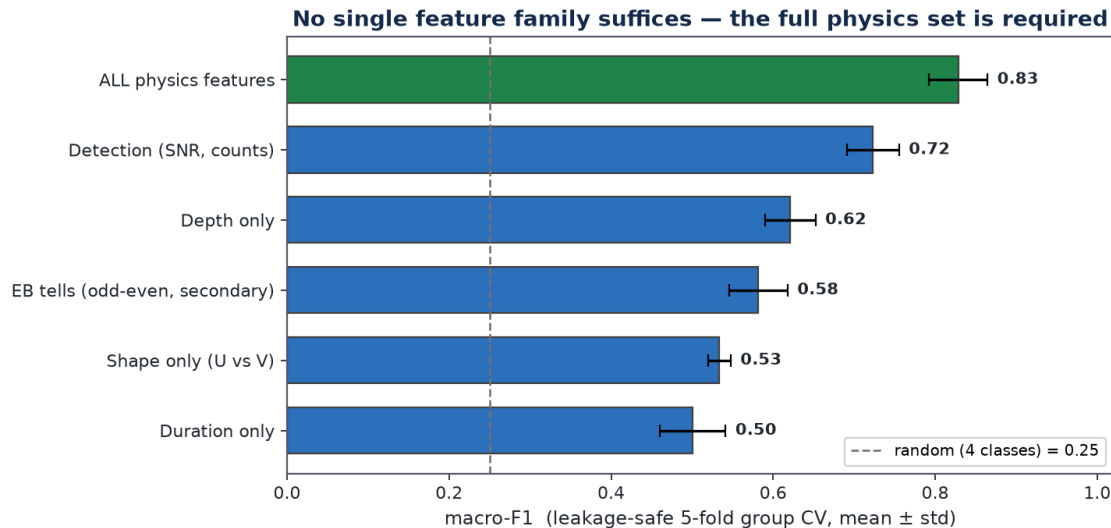


Left: TIC 100029948 — eclipsing binary flagged by physics features (depth ~25%, V-shape, odd–even 0.23).

Right: Pi Mensae c — blindly recovered ($P = 6.262$ d vs lit. 6.268; depth 289 ppm vs lit. ~315; SDE 12.3).

The detection spine works end-to-end today; the physics-branch classifier is validated next (CNN + curated labels in Round-2).

Feature ablation — no single feature suffices (synthetic labels)



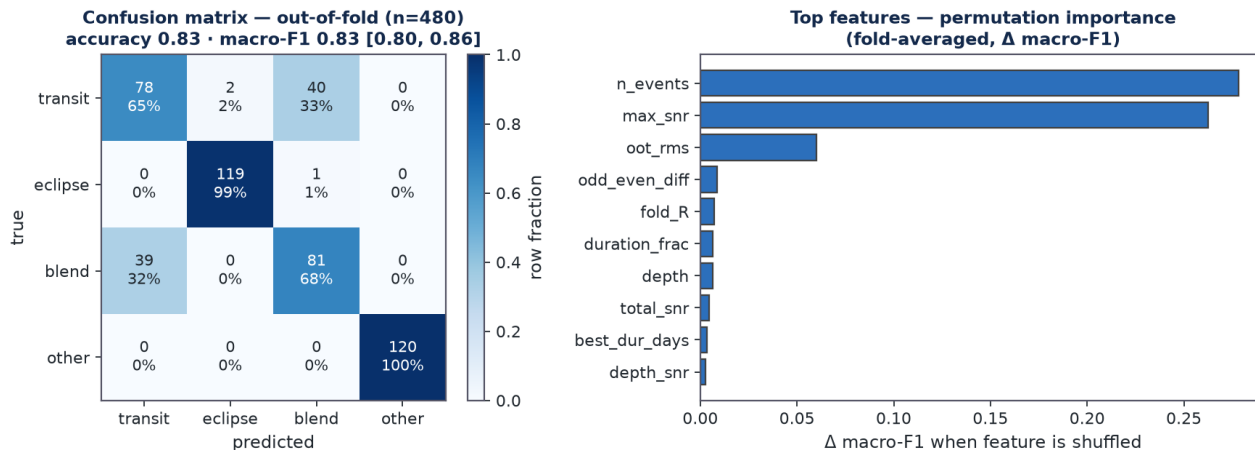
SYNTHETIC LABELS — injected signals on real TESS noise (proof of pipeline; not real-world accuracy)

Leakage-safe 5-fold group CV (no injection host shared train/test): depth-only 0.62, shape-only 0.53 — no single family exceeds 0.72; the full physics set reaches macro-F1 0.83.

Quantifies the committee's point — depth does not discriminate; the combined physics features do.

Classifier — leakage-safe proof of path (synthetic labels)

PS7 physics-branch classifier — leakage-safe evaluation

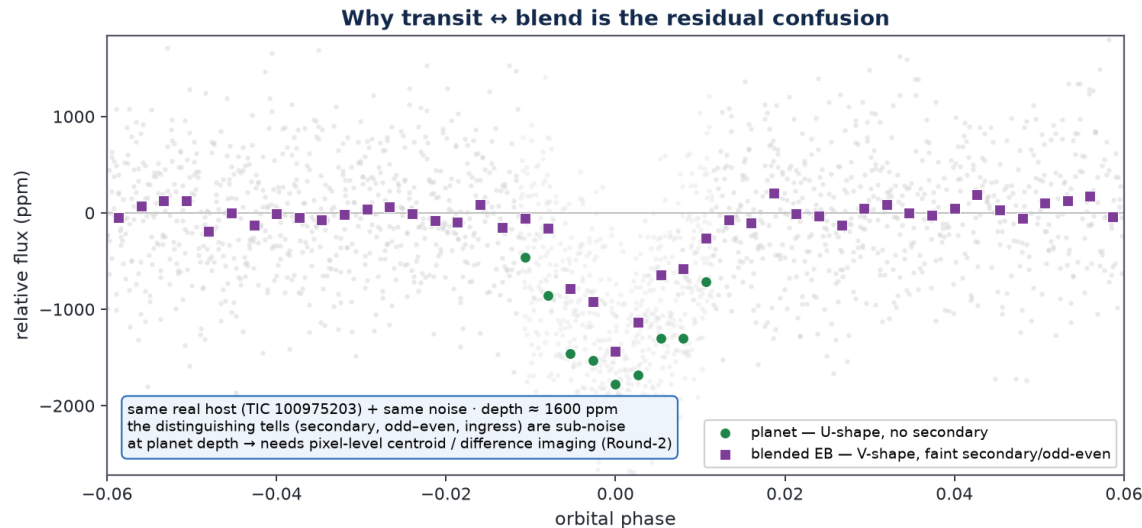


[SYNTHETIC LABELS — injected signals on real TESS noise (proof of pipeline; not real-world accuracy)]

Group-CV out-of-fold (n = 480): accuracy 0.83, macro-F1 0.83 (95% CI 0.80–0.86). Eclipses & systematics are separated cleanly; the residual is shallow transit vs blend.

Honest scope: synthetic labels validate the pipeline, not real-world accuracy — Round-2 uses the organisers' curated set + pixel-level features.

The hard case & Round-2 roadmap (transit vs blend)



SYNTHETIC LABELS — injected signals on real TESS noise (proof of pipeline; not real-world accuracy)

At planet depth, on the same real host + noise, a planet (U) and a blended EB (V + faint secondary/odd-even) are nearly identical from the light curve alone.

This is the residual confusion — and precisely why Round-2 adds pixel-level centroid / difference-imaging features.

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THANK YOU